

## PERSONAL DATA

Name: Vladimir R. Kostić  
Date of Birth: 6th January, 1980  
Place of Birth: Loznica, Serbia  
Nationality: Serbian



## CONTACT INFORMATION

Work Address: Computational Statistics & Machine Learning  
Center for Human Technologies  
Italian Institute of Technology  
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Web: https://vladi-slk.github.io  
Scholar: user=66gV7SAAAAAJ  
ORCID: 0000-0002-2876-8834

## LANGUAGE SKILLS

Serbian (native speaker), English (advanced C2),  
Italian (independent B1), French (beginner A2)

## CODING SKILLS

Python, MATLAB, R

## EDUCATION

2010 Ph.D. in Numerical Mathematics, Faculty of Science, University of Novi Sad, Serbia  
Theses "Benefits from Generalized Diagonal Dominance" supervised by Ljiljana  
Cvetković with external co-supervision by Richard S. Varga (Kent State University, USA)  
2009 M.Sc. in Numerical Mathematics, Faculty of Science,  
University of Novi Sad, Serbia  
2003 B.Sc.in Applied Mathematics, Faculty of Science, University of Novi Sad, Serbia

## HONORS AND AWARDS

2013 BMS Einstein Foundation Fellowship at the Technical University of Berlin  
2007 1st Young Scientists Award, MATTRIAD 2007 Conference, Będlewo, Poland  
2004 The Best Student of the University of Novi Sad Award, Serbia  
2004 The Honor Mileva Maric-Einstein of the University of Novi Sad  
2003 Fellowship of the Fund of the Royal House of Karađorđević  
2002 Award of High Academic Achievements by the  
Embassy of the Kingdom of Norway  
2001 – 2003 The Annual University Honors, University of Novi Sad (each year)

## TEACHING AND RESEARCH POSITIONS

|                   |                                                                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 09/2023 – present | Researcher in Machine Learning and Statistical Computing, Italian Institute of Technology, Genova, Italy                                  |
| 02/2020 – 08/2023 | Post-doctoral Researcher in Machine Learning and Statistical Computing, Italian Institute of Technology, Genova, Italy                    |
| 01/2016 – present | Associate Professor in Numerical Linear Algebra, University of Novi Sad, Serbia                                                           |
| 01/2011 – 01/2016 | Assistant Professor in Numerical Linear Algebra, University of Novi Sad, Serbia                                                           |
| 06/2013 – 02/2014 | Post-doctoral Researcher in Numerical Mathematics, Fellow of IMU Berlin Einstein Foundation Program, Technical University Berlin, Germany |
| 09/2004 – 12/2010 | Teaching Assistant, University of Novi Sad, Serbia                                                                                        |

## COURSES TAUGHT

Numerical Linear Algebra 1 & 2  
(MSc in Data Science and Applied Mathematics)  
Modeling of Dynamical Systems (BSc in Applied Mathematics)  
Scientific Computing (PhD in Mathematics)  
Software for Data Analysis (BSc in Biology and Ecology)  
Mathematical and Statistical Methods in Biological Research  
(PhD in Biology and Ecology)

## MENTORSHIP OF PHD THESES

|           |                                                                                                                                                                        |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025–     | Ilja Uzelac Bujišić, Faculty of Technical Sciences, University of Novi Sad, Serbia                                                                                     |
| 2013–2020 | Srdjan Milićević, PhD theses <i>Algorithms for computing the optimal Geršgorin-type localizations</i> at Faculty of Technical Sciences, University of Novi Sad, Serbia |
| 2013–2018 | Dragana Gardašević, PhD theses <i>Geršgorin-type localizations for Nonlinear Eigenvalue Problems</i> at Faculty of Technical Sciences, University of Novi Sad, Serbia  |
| 2014–2017 | Dragana Cvetković, PhD theses <i>New stability indicators for the empirical food webs</i> at Faculty of Technical Sciences, University of Novi Sad, Serbia             |

## INVITED LECTURES AND CONFERENCES

Since 2004 participated in over 50 international scientific conferences, organized 8 international scientific meetings and had over 20 invited scientific talks worldwide.

### INVITED LECTURES

|                |                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------|
| May 2026       | by Roberto Oliveira, IMPA, Rio de Janeiro, Brazil                                                   |
| April 2026     | by Stefano Serra Capizzano, University of Insubria, Como, Italy                                     |
| Januray 2026   | by Karim Lounici, SÃ©minaire Parisien de Statistique, IHP, Paris, France                            |
| June 2025      | by Arthur Gretton, Seminar of Gatsby Institute, London, UK                                          |
| June 2023      | by Patricia Reynaud-Bouret, UniversitÃ© CÃ´te d'Azur, Nice, France                                  |
| December 2022  | by Karim Lounici, Ãcole Polytechnique, Paris, France                                               |
| May 2018       | by Zoran TomljanoviÄ, University of Osijek, Croatia,                                               |
| September 2017 | by Agnieszka Miedlar, Kansas University, USA,                                                       |
| September 2015 | by Juan Manuel Pena, Univeristy of Zaragoza, Zaragoza, Spain                                        |
| August 2015    | by Yuji Nakatsukasa, Univeristy of Tokyo, Tokyo, Japan                                              |
| December 2014  | by Daniel Kressner, Ecole Polytechnique Federale de Lausanne, Switzerland                           |
| February 2014  | by Volker Mehrmann, Technical University, Berlin, Germany                                           |
| November 2013  | by Peter Benner, Max Planck Institute for Dynamics of Complex Technical Systems, Magdeburg, Germany |
| March 2009     | by Jaroslav ZemÃ¡nek, Banach Center, Warszawa, Poland                                               |

### INVITED CONFERENCE TALKS

|                       |                                                                                                                                                   |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| June 17-19, 2026      | Les journÃ©es de Statistiques du Sud, Nice, France                                                                                                |
| December 9-12, 2025   | General Conference of Future of AI Research (FAIR) Project, Rome, Italy                                                                           |
| July 8-12, 2025       | Neuromod Institute Annual Meeting, UniversitÃ© CÃ´te d'Azur, Antibes, France                                                                      |
| June 16-20, 2024      | New trends in statistical learning, Porquerolles, France                                                                                          |
| December 15, 2024     | Workshop on Data-driven modelling, analysis, and control using the Koopman operator, Conference on Decision and Control (CDC 2024), Milano, Italy |
| September 25-28, 2023 | Minisymposium on Data-driven dynamical systems, DMV, Ilmenau, Germany                                                                             |
| June 17-23, 2023      | New trends in statistical learning, Porquerolles, France                                                                                          |
| October 25-28, 2022   | MML, Bilbao, Spain                                                                                                                                |
| October 19-23, 2019   | NASC19, Nanjing, China,                                                                                                                           |
| July 18-22, 2016      | Minisymposium at 7ECM, Berlin, Germany,                                                                                                           |
| August 10-14, 2015    | Minisymposium ICIAM 2015, Beijing, China,                                                                                                         |
| March 23-27, 2009     | Conference MATTRIAD 2009 Conference, Będlewo, Poland                                                                                              |
| May 24-28, 2010       | Conference Applied Linear Algebra in Honor of Hans Schneider, Novi Sad, Serbia                                                                    |
| July 18-22, 2011      | Minisymposium MS380 at ICIAM 2011, Vancouver, BC, Canada                                                                                          |

## SHORT-TERM RESEARCH PROJECTS

- 08/2025            Operator learning for dynamical systems and rare events discovery, "Research program at CIRM", Luminy, France
- 01/2015            Matrix nearness problems for general Lyapunov-Type stability domains and their applications, "Research in Pairs" (RiP) at Mathematisches Forschungsinstitut Oberwolfach (MFO), Oberwolfach, Germany

## HEAD OF RESEARCH PROJECTS

- 2009 – 2015            Multilateral Cooperation in Numerical Linear Algebra  
Department of Mathematics, University of Ioannina, Greece,  
Faculty of Mathematics, Mechanics and Computer Science, Southern Federal University Rostov-on-Don, Russia,  
Department of Mathematics and Informatics, University of Novi Sad, Serbia. Grant of the Secretariat of Science and Technological Development of the Autonomous Province of Vojvodina
- 2025 – 2028            Spectral and Pseudo-spectral Estimation for Data-Driven Dynamical Systems (SPEEDY), University of Novi Sad, Serbia. Grant of the Secretariat of Science and Technological Development of the Autonomous Province of Vojvodina

## PARTICIPATION IN RESEARCH PROJECTS

- 2023–2025            Future Artificial Intelligence Research (FAIR),  
Project PE0000013 CUP J53C22003010006, NextGenerationEU initiative and the Italian National Recovery and Resilience Plan (PNRR) from the Ministry of University and Research (MUR)
- 2011 – 2020            Numerical Linear Algebra and Discrete Structures,  
Grant No. 174019, Ministry of Science and Technological Development of the Republic of Serbia
- 2011 – 2015            Applied Linear and Combinatorial Algebra,  
Grant of the Secretariat of Science and Technological Development of the Autonomous Province of Vojvodina
- 2008 – 2012            Multilateral Cooperation in Numerical Linear Algebra,  
The Institute for Computational Mathematics, Kent State University, USA, Department of Mathematics, University of Ioannina, Greece, Department of Mathematics and Informatics, University of Novi Sad, Serbia. Grant of the Secretariat of Science and Technological Development of the Autonomous Province of Vojvodina
- 2007 – 2012            Multilateral Cooperation in Applied Linear Algebra,  
Instituto de Matematica Multidisciplinar, Universidad Politecnica de Valencia, Spain, Faculty of Mathematics and Computer Science, Adam Mickiewicz University, Poland, Department of Mathematics and Informatics, University of Novi Sad, Serbia. Grant of the Secretariat of Science and Technological Development of the Autonomous Province of Vojvodina
- 2006 – 2010            Numerical Linear Algebra, Stochastic and Statistics with Applications,  
Grant No. 144025 of the Ministry of Science and Technological Development of the Republic of Serbia
- 2006 – 2007            Bilateral Cooperation in Numerical Linear Algebra  
Instituto de Matematica Multidisciplinar, Universidad Politecnica de Valencia, Spain, Department of Mathematics and Informatics, University of Novi Sad, Serbia. Grant of the Secretariat of Science and Technological Development of the Autonomous Province of Vojvodina
- 2005 – 2010            Tools of Linear and General Algebra in Computer Science,  
Grant of the Secretariat of Science and Technological Development of the Autonomous Province of Vojvodina

## CONFERENCE ORGANIZATION

|                        |                                                                                                                                                                                            |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| July 3–5, 2025         | Scientific and Organizing Committee, Applied Linear Algebra 2025 in honor of Zhong-Zhi Bai, Novi Sad, Serbia,                                                                              |
| June 30 – July 2, 2025 | Organizing Committee, 11th International Conference on Matrix Analysis and its Applications, Novi Sad, SERBIA                                                                              |
| September 16–20, 2013  | Secretary of the Organizing Committee, MATTRIAD 2013 Conference, Herceg Novi, Montenegro                                                                                                   |
| May 18–22, 2013        | Organizer of Minisimposia MA 2: Parameter depending matrix equations at 84th Annual Meeting of GAMM (The International Association of Applied Mathematics and Mechanics), Novi Sad, Serbia |
| May 18–22, 2013        | Secretary of the Local Organizing Team, 84th Annual Meeting of GAMM (The International Association of Applied Mathematics and Mechanics), Novi Sad, Serbia                                 |
| May 24–28, 2010        | Secretary of the Organizing Committee, Applied Linear Algebra in honor of Hans Schneider, Novi Sad, Serbia                                                                                 |
| April 28–30, 2008      | Secretary of the Organizing Committee, Applied Linear Algebra in honor of Ivo Marek, Novi Sad, Serbia                                                                                      |
| October 13–15, 2005    | Secretary of the Organizing Committee, Applied Linear Algebra in honor of Richard Varga, Palić, Serbia and Montenegro                                                                      |

## EDITORSHIP

|                   |                                                                                                                                                                        |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 06/2014 – 01/2017 | Newsletter of the European Mathematical Society, ISSN 1027-488X                                                                                                        |
| 12/2013           | PAMM, Wiley VCH, Online ISSN: 1617–7061 Special Issue: 84th Annual Meeting of the International Association of Applied Mathematics and Mechanics (GAMM), Novi Sad 2013 |

## REVIEWER

Proceedings of Machine Learning Research  
SIAM Journal on Optimization (SIOPT)  
SIAM Journal on Matrix Analysis and Applications (SIMAX)  
SIAM Journal on Mathematics of Data Science (SIMODS)  
Numerical Linear Algebra and Applications (NLAA)  
Linear Algebra and Applications (LAA)  
Electronic Transactions of Numerical Analysis (ETNA)  
Electronic Journal of Linear Algebra (ELA)  
Numerical Algorithms (NA)  
Applied Mathematics and Computation (AMC)  
Journal of Computational and Applied Mathematics (JCAM)  
Central European Journal of Mathematics (CEJM)  
Journal of Inequalities and Applications  
International Journal of Computer Mathematics

## MEMBERSHIP IN PHD THESES DEFENSE BOARDS:

|      |                                                                                                                                                                      |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2019 | Giulia Denevi, <i>Efficient Lifelong Learning Algorithms: Regret Bounds and Statistical Guarantees</i> at University of Genova, Italy                                |
| 2019 | Abear Saeed Aboglida Saeed, <i>Convergence analysis of modulus based methods for linear complementarity</i> at Faculty of Science, University of Novi Sad, Serbia    |
| 2017 | Nemanja Božović, <i>Deflation Based Krylov Subspace Methods for Sequences of Linear Systems</i> at Univeristy of Wuppertal, Germany                                  |
| 2016 | Maja Nedović, <i>The Schur complement and H-matrix theory</i> at Faculty of Technical Sciences, University of Novi Sad, Serbia                                       |
| 2014 | Ksenija Doroslovački, <i>Generalized diagonal dominance for block matrices and its applications</i> at Faculty of Technical Sciences, University of Novi Sad, Serbia |

## OTHER NGO ACTIVITIES

|                   |                                                                                                                                                                                                       |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03/2018 –11/2018  | Trainer of Qualification course "Building Peace from the inside out" with Berghof Foundation, Tübingen and Relief International in Azraq and Zaatari Refugee Camps in Jordan                          |
| 06/2018           | Trainer of Qualification course "Conflict sensitive communication in work with migrants" with International Organization for Migration for workers in Commissariat for Refugees of Republic of Serbia |
| 03/2017 –09/2017  | Trainer of ToT "Building Peace from the inside out" with Berghof Foundation, Tübingen and Relief International in Azraq and Zaatari Refugee Camps in Jordan                                           |
| 05/2017 –06/2017  | Trainer of "Coaching for Managers" in Coca-Cola Company, Belgrade, Serbia, with Unique Ideas, Belgrade, Serbia                                                                                        |
| 10/2016 –11/2016  | Trainer of Qualification course "Building Peace from within" with Berghof Foundation, Tübingen and Relief International in Azraq and Zaatari Refugee Camps in Jordan,                                 |
| 10/2014 – 06/2019 | Consultant and Trainer for Berghof Foundation, Germany, Project Nonviolent and Civic Education in Jordan                                                                                              |

## SELECTED PUBLICATIONS

Published over 40 scientific papers in top journals on SCI list and more than 15 papers in top conference proceedings.

- T. Germain, R. Flamary, **V.R. Kostic** and K. Lounici. *A Spectral-Grassmann Wasserstein metric for operator representations of dynamical systems*. In The Fourteenth International Conference on Learning Representations (ICLR) (2026)
- T. Devergne, **V. Kostic**, M. Pontil, M. Parrinello. *The seeds of the future are in the present: A blind exploration of metastable states*. Proceedings of the National Academy of Sciences, 123(5) (2026)
- T. Devergne, **V.R. Kostic**, M. Pontil and M. Parrinello. Slow dynamical modes from static averages. The Journal of Chemical Physics. Vol. 162 No. 12 (2025)
- **V. Kostić**, K. Lounici, H. Halconruy, T. Devergne and M. Pontil. *Learning the Infinitesimal Generator of Stochastic Diffusion Processes*. Advances in Neural Information Processing Systems 37 (NeurIPS) (2024)
- **V. Kostić**, K. Lounici, G. Pacreau, P. Novelli, G. Turri and M. Pontil *Neural Conditional Probability for Inference*. Advances in Neural Information Processing Systems 37 (NeurIPS2024)
- **V.R. Kostić**, P. Inzerili, K. Lounici, P. Novelli and M. Pontil. *Consistent Long-Term Forecasting of Ergodic Dynamical Systems*. In 41st International Conference on Machine Learning (ICML) (2024)
- **V.R. Kostic**, P. Novelli, R. Grazzi, K. Lounici, and M. Pontil. *Learning invariant representations of time-homogeneous stochastic dynamical systems*. In The Twelfth International Conference on Learning Representations (ICLR) (2024)
- **V.R. Kostić**, K. Lounici, P. Novelli and M. Pontil, *Sharp Spectral Rates for Koopman Operator Learning*, Advances in Neural Information Processing Systems 36 (NeurIPS2023 spotlight) (2023)
- **V.R. Kostić**, P. Novelli, A. Maurer, C. Ciliberto, L. Rosasco and M. Pontil, *Learning Dynamical Systems via Koopman Operator Regression in Reproducing Kernel Hilbert Spaces*, Advances in Neural Information Processing Systems 35 (NeurIPS) (2022)
- **V.R. Kostić** and S. Salzo, *The method of randomized Bregman projections for stochastic feasibility problems*. Numerical Algorithms 93 (3), pp. 1269–1307 (2022)
- **V.R. Kostić**, Lj. Cvetković and D. Lj. Cvetković, *Pseudospectra localizations and their applications*, Numerical Linear Algebra with Applications 23 (2), pp. 356–372 (2016)
- **V.R. Kostić**, *On general principles of eigenvalue localizations via diagonal dominance*, Advances in Computational Mathematics No. 41, pp. 55–75 (2015)
- Lj. Cvetković, **V.R. Kostić** and R. S. Varga, *A new Geršgorin-type eigenvalue inclusion set*, ETNA Elec. Trans. on Numer. Anal., No. 18, pp. 73– 80 (2004).

On September 19, 2026 according to Google Scholar publications have **1570 citatons** (893 since 2020), **h-index is 22** (14 since 2021) and **i10-index is 31** (19 since 2021).

## BOOKS &amp; MONOGRAPHS

- B1. Cvetković, Lj. and V. Kostić, *Collection of Solved problems in Numerical Linear Algebra* (in serbian), University of Technical Sciences, University of Novi Sad, Serbia (2015)
- B2. Lj. Cvetković, Bjelica, D., V. Kostić, and Kovačević, M., *Matrično-verovatnosni modeli u sportu*, Crnogorska Sportska Akademija, Podgorica (2005)

## JOURNAL PUBLICATIONS

- J1. Kostic, V.R., Lounici, K., Pontil, M., *Toeplitz Based Spectral Methods for Data-driven Dynamical Systems*. Numerical Linear Algebra and Applications (2026)
- J2. Turri, G., Pacreau, G., Meanti, G., Devergne, T., Ordonez, D., Mirzaei, E., Belucci, B., Lounici, K., Kostic, V., Pontil, M. and Novelli, P., *kooplearn: A Scikit-Learn Compatible Library of Algorithms for Evolution Operator Learning*. Journal of Open Source Software, 11(122), 10342, (2026)
- J3. T. Devergne, V. Kostic, M. Pontil, M. Parrinello. The seeds of the future are in the present: A blind exploration of metastable states. Proceedings of the National Academy of Sciences, 123(5), e2524602123 (2026)
- J4. G. Turri, V. R. Kostic, P. Novelli, M. Pontil. A randomized algorithm to solve reduced rank operator regression. SIAM Journal on Mathematics of Data Science, 8(1), 23-45 (2026)
- J5. D. Ordoñez-Apraéz, G. Turrise, V.R. Kostic, M. Martin, A. Agudo, F. Moreno-Noguer, M. Pontil, C. Semini and C. Mastalli. Morphological Symmetries in Robotics. The International Journal of Robotics Research, 02783649241282422 (2025)
- J6. T. Devergne, V. R. Kostic, M. Pontil and M. Parrinello. Slow dynamical modes from static averages. The Journal of Chemical Physics. Vol. 162 No. 12 (2025)
- J7. V. Kostić and S. Salzo, *The method of randomized Bregman projections for stochastic feasibility problems*. Numerical Algorithms 93 (3), pp. 1269–1307 (2022)
- J8. V. Kostić, L. Cvetković, E. Šanca, *From pseudospectra of diagonal blocks to pseudospectrum of a full matrix*, Journal of Computational and Applied Mathematics, Vol 386, online 113265, (2020)
- J9. S. Milićević, V. Kostić, Lj. Cvetković and A. Międlar, *An implicit algorithm for computing the minimal Geršgorin set*, Filomat, Vol. 33, No 13, pp. 4229–4238, (2019)
- J10. V. Kostić., D. Gardašević, *On the Geršgorin-type localizations for nonlinear eigenvalue problems*, Applied Mathematics and Computation, Vol. 337, No 1, pp. 179–189, (2018)
- J11. Lj. Cvetković, V. Kostić, E. Šanca, *Fractional pseudospectra and their localizations*, Linear Algebra and Its Applications, No. 559, pp. 244–269, (2018)
- J12. Lj. Cvetković, V. Kostić, E. Šanca, A. Saed, *Error Control Based on the Novel Proof of Convergence of the MSMAOR Methods for the LCP*, East Asian Journal on Applied Mathematics, Vol. 8, No. 2, pp. 352–364 (2018)
- J13. V. Kostić and Lj. Cvetković, *On the inertia of the block H-matrices*, Numerical Linear Algebra and its Applications No. 24(5), pp. e2101 (2017)
- J14. D. Mihailović, V. Kostić, I. Balaž and D. Kapor, *Computing the threshold of the influence of intercellular nanotubes on cell-to-cell communication integrity*, Chaos, Solitons & Fractals No. 91, pp. 174–179 (2016)

- J15. V. Kostić, Lj. Cvetković, K. Doroslovački and D. Cvetković, *Euclidean norm estimates of the inverse of some special block matrices*, Applied Mathematics and Computation No. 284, pp. 12–23 (2016)
- J16. V. Kostić, A. Międlar and Lj. Cvetković, *An algorithm for computing minimal Geršgorin sets*, Numerical Linear Algebra and its Applications no. 23(2), pp. 272–290 (2016)
- J17. V. Kostić, Lj. Cvetković and D. Cvetković, *Pseudospectra localizations and their applications* Numerical Linear Algebra and its Applications no. 23(2), pp. 356–372 (2016)
- J18. V. Kostić, Lj. Cvetković and D. Cvetković, *Improved stability indicators for empirical food webs*, Ecological modelling No. 320, pp. 1–8 (2016)
- J19. Lj. Cvetković, V. Kostić and E. Šanca, *A wider convergence area for the MSTMAOR iteration methods for LCP*, Numerical Algorithms No. 71 (1), pp. 77–88 (2016)
- J20. D. Mihailović, V. Kostić, G. Mimić and Lj. Cvetković, *Stability analysis of turbulent heat exchange over the heterogeneous environmental interface in climate models*, Applied Mathematics and Applications No. 265, pp. 79–90 (2015)
- J21. V. Kostić, A. Międlar and J. Stolwijk, *On matrix nearness problems: distance to delocalization*, SIAM J. Matrix Anal. & Appl., No. 36(2), pp. 435–460 (2015)
- J22. V. Kostić, *On general principles of eigenvalue localizations via diagonal dominance*, Advances in Computational Mathematics No. 41, pp. 55–75 (2015)
- J23. Lj. Cvetković, V. Kostić and M. Nedović, *Generalizations of Nekrasov matrices and applications*, Open Mathematics No. 13(1), DOI: 10.1515/math-2015-0012 (2015)
- J24. J. Aleksić, V. Kostić and M. Žigić, *Spectrum localizations for matrix operators on  $\ell^p$  spaces*, Applied Mathematics and Computation No. 249, pp. 541–553 (2014)
- J25. Lj. Cvetković, A. Hadjidimos and V. Kostić, *On the choice of parameters in MAOR type splitting methods for the linear complementarity problem*, Numer. Algorithms No. 67(4), pp. 793–806 (2014)
- J26. D. T. Mihailović, V. Kostić, I. Balaž and Lj. Cvetković, *Complexity and asymptotic stability in the process of biochemical substance exchange in a coupled ring of cells*, Chaos Fractals and Solitons, No. 65, pp. 30–43 (2014)
- J27. Lj. Cvetković and V. Kostić, *A note on the convergence of the MSMAOR method for linear complementarity problems*, Numer. Lin. Alg. Appl., No. 9 (4), pp. 534–539 (2014)
- J28. Lj. Cvetković, V. Kostić and K. Doroslovački, *Max-norm bounds for the inverse of S-Nekrasov matrices*, Appl. Math. Comput., No. 218 (18), pp. 9498–9503 (2012)
- J29. Lj. Cvetković and V. Kostić, *Application of Generalized Diagonal Dominance in Wireless Sensor Network Optimization Problems*, Appl. Math. Comput. No. 218, pp. 4798–4805 (2012)
- J30. V. Kostić, R. S. Varga and Lj. Cvetković, *Localization of Generalized Eigenvalues by Cartesian Ovals*, Numer, Linear Algebra Appl., No. 19 (4), pp. 728–741 (2012)
- J31. Lj. Cvetković, V. Kostić and J. M. Pena, *Eigenvalue localization refinements related to positivity*, SIAM J. Matrix Anal. Appl., No. 32 (3), pp. 771–784 (2011)
- J32. Lj. Cvetković, R. Bru, V. Kostić, and F. Pedroche, *A simple generalization of Geršgorin's theorem*, Adv. Comput. Math., No. 35, pp. 271–280 (2011)
- J33. R. Bru, Lj. Cvetković, V. Kostić and F. Pedroche, *Characterization of  $\alpha_1$  and  $\alpha_2$  matrices*, Cent. Eur. J. Math., No. 8 (1), pp. 32–40 (2010)
- J34. R. Bru, Lj. Cvetković, V. Kostić, and F. Pedroche, *Sums of  $\Sigma$ -strictly diagonally dominant matrices*, Linear and Multilinear Algebra, No. 58 (1), pp. 75–78 (2010)
- J35. V. Kostić, Lj. Cvetković and R. S. Varga, *Geršgorin-type localizations of generalized eigenvalues*, Numer. Linear Algebra Appl., No. 16 (11-12), pp. 883–898 (2009)

- J36. Lj. Cvetković, V. Kostić and S. Rauški, *A new subclass of  $H$ -matrices*, Appl. Math. Comput., No. 208, pp. 206–210 (2009)
- J37. R. S. Varga, Lj. Cvetković and V. Kostić, *Approximation of the minimal Geršgorin set of a square complex matrix*, ETNA Elec. Trans. on Numer. Anal., No. 30, pp. 398–405 (2008)
- J38. Lj. Cvetković, V. Kostić, M. Kovačević and T. Szulc, *Further results on  $H$ -matrices and their Schur complements*, Appl. Math. Comput., No. 198(2), pp. 506–510 (2008)
- J39. Lj. Cvetković and V. Kostić, *A note on the convergence of the AOR method*, Appl. Math. Comput. No. 194(2), pp. 394–399 (2007)
- J40. Lj. Cvetković and V. Kostić, *New subclasses of block  $H$ -matrices with applications to parallel decomposition-type relaxation methods*, Numer. Algorithms, No. 42 (3-4), pp. 325–334 (2006)
- J41. Lj. Cvetković and V. Kostić, *Between Geršgorin and minimal Geršgorin sets*, J. Comput. Appl. Math., No. 196 (2), pp. 452–458. (2005)
- J42. Lj. Cvetković and V. Kostić, *New criteria for identifying  $H$ -matrices*, J. Comput. Appl. Math., No. 180 (2), 265–278 (2005)
- J43. Lj. Cvetković, V. Kostić, and R. S. Varga, *A new Geršgorin-type eigenvalue inclusion set*, ETNA Elec. Trans. on Numer. Anal., No. 18, pp. 73–80 (2004)
- J44. Lj. Cvetković, V. Kostić and M. Kovačević, *On Geršgorin-type eigenvalue inclusion theorems*, Bull. Appl. Math., No. 2018 (C), pp. 351–358 (2003)

## PROCEEDINGS PUBLICATIONS

- P1. Frohlich, A., Kostic, V., Lounici, K., Perazzo, D. and Pontil, M., *Toward Scalable and Valid Conditional Independence Testing with Spectral Representations*. Proceedings of the 43rd International Conference on Machine Learning (ICML2026)
- P2. Meunier, D., Wornbard, J., Kostic, V.R., Moulin, A., Fröhlich, A., Lounici, K., Pontil, M. and Gretton, A., *Outcome-Aware Spectral Feature Learning for Instrumental Variable Regression*. Proceedings of the 43rd International Conference on Machine Learning (ICML2026)
- P3. Ordonez-Apraez, D., Kostic, V., Fröhlich, A., Brandt, V., Lounici, K. and Pontil, M., *Equivariant Representation Learning for Symmetry-Aware Inference with Guarantees*. Proceedings of the 43rd International Conference on Machine Learning (ICML2026)
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## CURRENT SUBMISSIONS

- S1. Kostic, V.R., Cvetkovic, D. Lj, Cvetkovic, Lj. *Scalable Pseudospectral Analysis via Low-Rank Approximations of Dynamical Systems*. arXiv preprint arXiv:2602.01721 (2026). [submitted to JCAM]
- S2. Kostic, V.R., Lounici, K., Halconruy, H., Devergne, T., Parrinello, M. and Pontil, M., *Langevin-Informed Transfer Learning: Replacing the Target Samples by Black-box Feedback*. arXiv preprint (2026) [submitted to NeurIPS]
- S3. Belucci, B., Lounici, K., Kostic, V.R. and Meziani, K., *VertCoHiRF: Decentralized Vertical Clustering Beyond k-means*. arXiv preprint arXiv:2602.07279. (2026) [submitted to NeurIPS]
- S4. Meziani, K., Belucci, B., Lounici, K. and Kostic, V.R., *CoHiRF: Hierarchical Consensus for Interpretable Clustering Beyond Scalability Limits*. arXiv preprint arXiv: 2502.00380. (2025)

## CONFERENCE TALKS

- T1. V. Kostić, S. Salzo, M. Pontil, *Linear Convergence of Batch Greenkhorn for Regularized Multimarginal Optimal Transport*, NeurIPS 2021 Workshop Optimal Transport and Machine Learning, Virtual-only, December 6-14, 2021
- T2. V. Kostić, *M-matrices as a Tool for Spectral and Pseudospectral Analysis*, NASC 2019, Nanjing, China, October 19–23, 2019
- T3. V. V. Kostić, *On pseudospectra localizations and their applications*, 7ECM 2016, Berlin, Germany, July 18–22, 2016.
- T4. V. Kostić, *Efficient computation of the minimal Geršgorin set*, ILAS 2016, Leuven, Belgium, July 11–15, 2016.
- T5. V. Kostić, *Matrix Nearness Problems for General Lyapunov-Type Stability Domains*, ICIAM 2015, Beijing, China, August 10–14, 2015.
- T6. V. Kostić, *Matrix Nearness Problems for General Lyapunov-Type Stability Domains*, SIAM LA 2015, Atlanta, Georgia, USA, June 26–30, 2015.
- T7. V. Kostić, *Generalized diagonal dominance in wireless sensor network optimization*, MSCx 2014, Salina, Sicily, Italy, June 09–13, 2014.
- T8. J. Aleksić, J., V. Kostić, M. Žigić, *Spectrum localizations for matrix operators on  $l^p$  spaces*, MATTRIAD 2013, Herceg Novi, Montenegro, September 16–20, 2013.
- T9. V. Kostić, Lj. Cvetković, Ė. Sanca, *Diagonal dominance in Euclidean norm*, MATTRIAD 2013, Herceg Novi, Montenegro, September 16–20, 2013.
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- T11. V. Kostić, *Application of M-matrices in Neural Network Models of Brain Activity*, Numerical Algebra and Scientific Computing, Dalian, China, January 20-24, 2013.
- T12. V. Kostić, Lj. Cvetković, *On stable formation of cortical cognitive maps via M-matrices*, 5th Conference on Numerical Analysis (NumAn 2012) - Recent Approaches to Numerical Analysis: Theory, Methods and Applications, Ioannina, Greece, September 5-8, 2012.
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- T19. Lj. Cvetković, V. Kostić, S. Rauški, *A new subclass of H-matrices*, GAMM 79th Annual, Bremen, Germany, March 31 – April 4, 2008.
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- T33. Lj. Cvetković, V. Kostić, *Some new characterizations of H-matrices*, 11th Conference of the International Linear Algebra Society (ILAS), Coimbra, Portugal, July 19–22, 2004.
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## TEACHING AND TRAINING MANUALS

Nolden D., Kostić V., *Strengthening Resilience - Building Peace from within* (in english and arabic), Berghof Foundation, Tübingen, Germany (2017)